# Counting Oriented Cycles Up to Dihedral Symmetry

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**Abstract**

I count directed cycles on $n$ edges under dihedral symmetry using a small transfer matrix coupled with Burnside's lemma. The adjacency rule forbids like-signed nonzero vertex signatures from being adjacent. A 6-state transfer matrix produces closed-walk counts; Burnside aggregates rotation and reflection fixed points to yield exact orbit counts. A full example for $n = 8$ reproduces the program values 288, 64, and 22. Optional primitivity and forbidden-subsequence filters follow immediately. Small $n$ results match brute force and OEIS A053656.

## 1 Model

A cycle is encoded by a binary edge sequence $(e_0, \ldots, e_{n-1}) \in \{0,1\}^n$. Edge $i$ connects vertices $i$ and $i+1 \pmod{n}$. Bit 0 orients clockwise, bit 1 counterclockwise. At vertex $v$ I define the signature

$$\sigma(v) = 2 \cdot (\text{incoming edges at } v) - 2 \in \{-2, 0, +2\}.$$

**Adjacency rule.** Nonzero signatures of the same sign may not be adjacent around the cycle.

## 2 Symmetry

The dihedral group $D_n$ ($n$ rotations, $n$ reflections; size $2n$) acts on edge sequences. A reflection reverses order and flips bits $0 \leftrightarrow 1$ to preserve geometric direction. Two sequences are equivalent if related by an element of $D_n$. Counting distinct cycles means counting orbits of this action.

## 3 Transfer matrix

Local state $s = (v, e)$ records previous vertex signature $v \in \{-2, 0, +2\}$ and current edge bit $e \in \{0, 1\}$; six states total. Given current edge $e$ and next edge $e'$, the next signature is

$$\sigma' = 2(\mathbf{1}_{e=0} + \mathbf{1}_{e'=1}) - 2.$$

I admit a transition $(v, e) \to (\sigma', e')$ only if the adjacency rule holds for $v, \sigma'$. The resulting $6 \times 6$ matrix $M$ has entries 1 for allowed transitions, 0 otherwise. The number of valid length-$m$ closed walks is

$$C(m) = \mathrm{tr}(M^m).$$

# 4 Burnside aggregation

**Rotations.** A rotation by $k$ fixes sequences with period $d = \gcd(n, k)$. Grouping rotations by $d$ yields

$$\mathrm{RotSum}(n) = \sum_{d|n} \varphi\left(\tfrac{n}{d}\right) C(d),$$

with Euler totient $\varphi$.

**Reflections.** For even $n$, a half-length transfer with symmetry boundary conditions gives $R(n)$, the number of reflection-fixed sequences; for odd $n$, $R(n) = 0$. The reflection contribution is $\frac{n}{2} R(n)$.

**Orbit count.**

$$a(n) = \frac{\mathrm{RotSum}(n) + \mathrm{RefTerm}(n)}{2n}, \qquad \mathrm{RefTerm}(n) = \tfrac{n}{2} R(n).$$

# 5 Worked example: $n = 8$

Divisors: $1, 2, 4, 8$. Transfer powers give

$$C(1) = 2, \quad C(2) = 4, \quad C(4) = 16, \quad C(8) = 256.$$

Weighted by totient:

$$\mathrm{RotSum}(8) = 4 \cdot 2 + 2 \cdot 4 + 1 \cdot 16 + 1 \cdot 256 = 288.$$

Reflection routine: $R(8) = 16$, so $\mathrm{RefTerm}(8) = 4 \times 16 = 64$. Therefore

$$a(8) = \tfrac{288+64}{16} = 22.$$

# 6 Optional filters

**Primitivity.** Using Möbius inversion,

$$a_{\mathrm{prim}}(n) = \sum_{d|n} \mu(d)\, a\left(\tfrac{n}{d}\right).$$

**Forbidden subsequences.** Canonical signatures from smaller $n$ may be declared forbidden; a length-$n$ signature is rejected if any cyclic substring matches a forbidden canonical representative. This path enumerates signatures directly (used for small $n$).

## 7 Validation and cost

For $n \leq 15$ I compare Burnside+transfer counts with brute-force canonical enumeration; they agree. Unconstrained counts match OEIS A053656 on the tested range. Runtime is driven by divisor iteration and fast exponentiation of the fixed 6-state matrix, so million-scale $n$ is practical.

## 8 Outlook

Next steps include asymptotic constants under additional local rules and extending the forbidden-subsequence filter with transfer-based acceleration.

## 9 Connection to the six-vertex model

The SIAM News article on *active hydraulics* describes a square grid of pipes where the flow at each junction satisfies a local conservation rule: two streams enter and two leave, so each junction has exactly six admissible arrow patterns [2, 3]. This is the classical *six-vertex (square ice) model* introduced to idealize hydrogen bonding in ice [5] and solved via transfer-matrix methods in statistical mechanics [4, 6, 1].

My cyclic-counting project is not the six-vertex model, but the methodology lines up closely:

- In both settings, the global object is built from *local compatibility rules*. In my case the adjacency constraint is checked from consecutive vertex signatures; in the six-vertex model the ice rule is checked at each vertex.
- In both settings, a *transfer matrix* packages local rules into a finite-state automaton. For my cycle it is a $6 \times 6$ matrix for allowed local steps; in the six-vertex model one uses a row-to-row transfer matrix to compute the partition function.
- In both settings, periodic boundary conditions naturally lead to $\mathrm{tr}(M^n)$-type expressions (counting closed walks, or counting configurations on a ring/torus).
- The *symmetry quotient* I take with Burnside (dihedral rotations/reflections of a cycle) parallels the general idea of identifying physically/structurally equivalent configurations. In the active-hydraulics experiment, symmetries also pair junction types (mirror-related vertices) and help explain which local states are energetically or statistically equivalent [2].

This connection is useful for exposition: it justifies why transfer matrices are a standard tool for lattice counting problems, and it frames Burnside's lemma as the "group-quotient layer" that sits on top of a transfer-matrix count when we also want to identify configurations under rotation/reflection.

# 10 Real-world use case (how this solves a design question)

In a lab or device context, one often wants to know how many *distinct* steady patterns a symmetric ring-like network can realize under a local rule. For example, consider an $n$-module circular device whose local module rule is encoded by an adjacency constraint (preventing two "same-sign" active junctions in a row) and whose physical symmetries identify configurations under rotation and flip of the ring. My computation answers:

- *How many admissible global configurations exist?* (transfer-matrix closed-walk count)
- *How many of those are genuinely different once the device symmetries are taken into account?* (Burnside orbit count)

This is immediately practical for experimental design (how many nonredundant states to test) and for information-capacity estimates (how many unique steady configurations the device can represent).

# References


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